

MUSKAN PERIWAL

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SUMMARY

Aspiring **AI/ML and Full-Stack Software Engineer** with hands-on experience developing scalable web applications, medical imaging platforms, and AI-powered solutions using Python, Flask, React, and SQL. Experienced in REST API development, medical image visualization (DICOM/NIfTI), backend engineering, and integrating AI/ML workflows into production applications. Passionate about building scalable software, solving complex engineering problems, and applying machine learning to real-world healthcare applications.

EDUCATION

Indian Institute of Technology, Madras B.S. in Data Science and Applications	<i>Chennai, Tamil Nadu</i> <i>Sep. 2022 – Present</i>
Maharaja Ganga Singh University B.Sc.	<i>Suratgarh, Rajasthan</i> <i>Aug. 2019 – Mar. 2022</i>
Devendar Baal Bharti Sr. Sec. School 12th – Non-Medical	<i>Sri Vijaynagar, Rajasthan</i> <i>Apr. 2018 – Mar. 2019</i>

WORK EXPERIENCE

Sudha Gopalakrishnan Brain Centre, IIT Madras Full Stack Developer Intern	<i>Chennai, Tamil Nadu</i> <i>Feb. 2026 – Present</i>
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- Developed a real-time scientific imaging viewer for monitoring cryosection and Huron scanning systems using Flask and JavaScript.
- Implemented live APIs, TIFF image processing, stateful machine tracking, and optimized dashboard rendering for continuous scan monitoring.
- Built BFI image visualization features for the Apollo Human Brain Workbench, enabling side-by-side raw/processed scan comparison and HD TIFF rendering for large-scale neuroscience datasets.
- Integrated OpenLayers and IIPImage tile-based viewers for efficient rendering of high-resolution brain histology images using dynamic JP2/TIFF switching.
- Built a full-stack pre-registration and collaborator onboarding feature in OpenAtlas using React, TypeScript, and Django REST Framework, enabling admins to invite external users before account creation with automatic project access assignment after signup.
- Developed secure invitation APIs, email automation, validation handling, role-based access control, and an admin management UI for tracking and managing pending collaborator invitations.
- Developed medical imaging capabilities for the OHIF Viewer by integrating NIfTI (.nii/.nii.gz) support, enabling browser-based MRI visualization and preparing the platform for AI-assisted analysis workflows.
- Extended Cornerstone3D imaging tools including measurements, segmentation, annotations, zoom, and MPR to support NIfTI datasets for advanced medical image analysis.
- Developed an AI-based MRI similarity search pipeline using medical vision foundation models to generate embeddings for retrieval of visually similar brain scans.

PROJECTS

MRI Similarity Search Integrated with OHIF Python, PyTorch, Hugging Face, Medical Imaging	<i>July. 2026</i>
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- Built an embedding-based medical image retrieval system for MRI scans.
- Generated embeddings using pretrained medical vision foundation models.
- Compared scans using cosine similarity.
- Designed the pipeline for scalable medical image search.

- Developed a web application with secure role-based access for Admin, Doctor, and Patient roles.
- Implemented authentication, authorization, and session management with Flask-Login.
- Implemented patient records management, appointment booking, doctor dashboards, and soft delete functionality.
- Designed the database using SQLAlchemy ORM and implemented full CRUD operations.
- Used MVC architecture and Jinja2 templates for dynamic UI rendering.

- Developed a full-stack web application to manage and track user skills with authentication support.
- Built RESTful APIs using Flask for user registration, login, and skill CRUD operations.
- Designed and implemented relational database models using SQLAlchemy.
- Implemented protected dashboard views for authenticated users.
- Followed MVC architecture and used React.js for dynamic UI rendering.

TECHNICAL SKILLS

Programming	Python, SQL, JavaScript
Frontend	React.js, HTML, CSS
Backend	Flask, FastAPI, Django REST Framework, REST APIs
AI/ML	PyTorch, Transformers, Hugging Face, NumPy, Pandas
Databases	SQLite, PostgreSQL, CRUD Operations, SQLAlchemy ORM
Tools & Platforms	Git, GitHub, Postman, VS Code
Libraries & Frameworks	SQLAlchemy, Jinja2, OpenLayers, OHIF, Cornerstone3D
Core Concepts	API Integration, Role-Based Access Control (RBAC), Debugging, Problem Solving